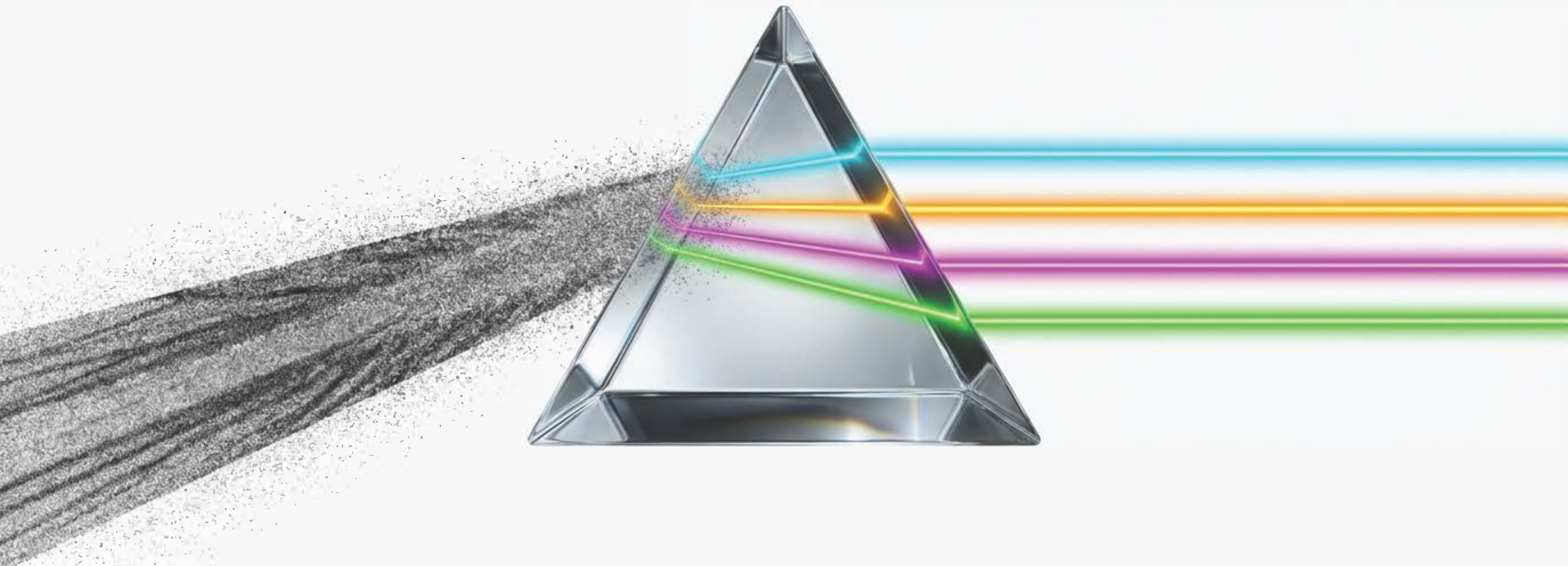
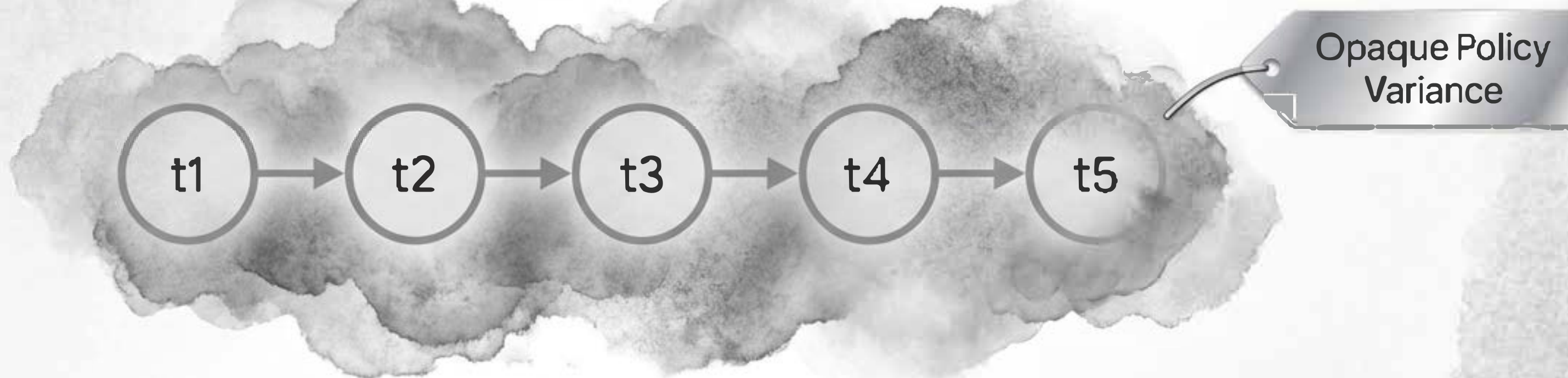


The Interpretable Prism

Decomposing Epistemic Uncertainty in Generative
Flow Networks via Polynomial Chaos Surrogates



Sequential generation inherits opaque uncertainty from imperfect reward models.



The Goal

Construct a sequence (molecule, reaction, reasoning chain) step-by-step.

The Bottleneck

The reward function relies on neural proxies trained on limited, noisy experimental data.

The Unknown

Total variance is measurable, but which specific step is fragile, and why?

"Existing methods tell us *that* a generative policy is uncertain, but not *why* or *which components* of the input drive that uncertainty."

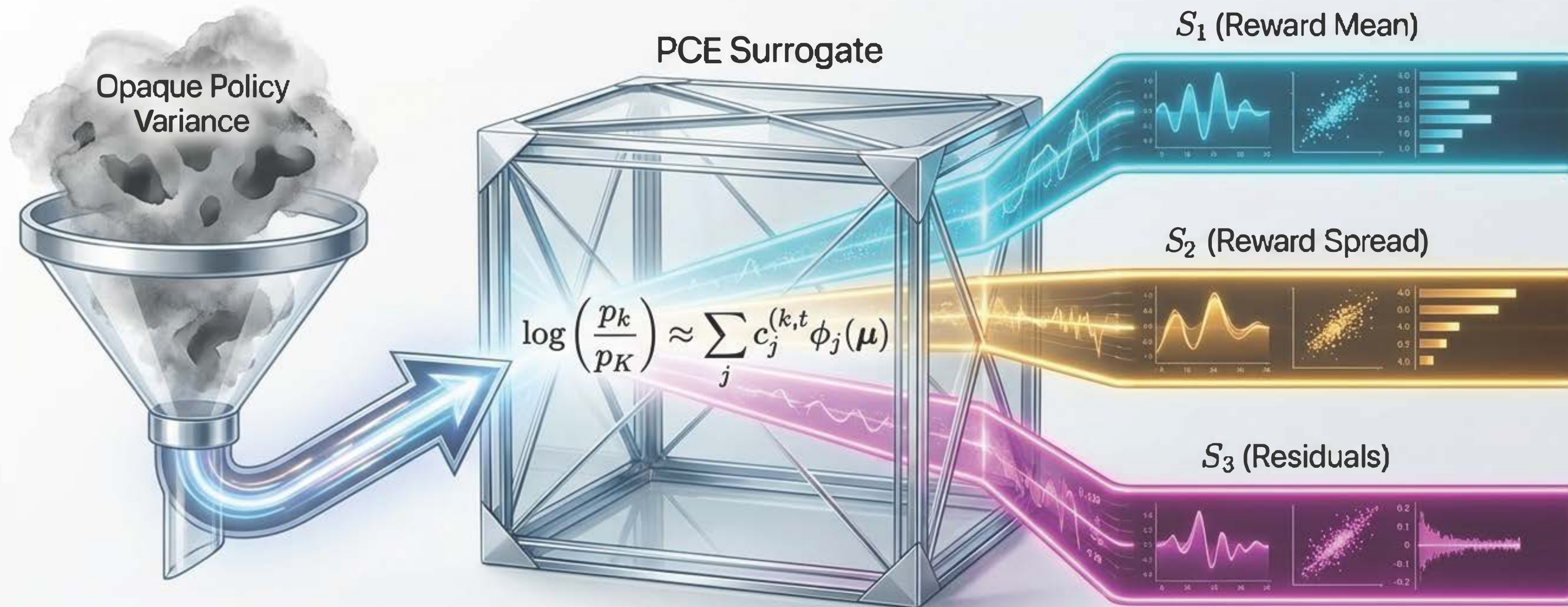
Deep ensembles quantify variance, but fail to explain its structural causes.

	Deep Ensembles	Gaussian Processes (GP)	Multilayer Perceptrons (MLP)	PCE Surrogate (Ours)
Provides Uncertainty?	✓ Yes	✓ Yes	✗ No	✓ Yes
Analytical Sobol' Indices?	✗ No	✗ No (Requires expensive MC)	✗ No	✓ Yes (Closed Form)
Fit Time	Retraining required	36.6s	0.68s	0.026s
Mean Absolute Error	Baseline	0.017	0.018	0.016

Only PCE surrogates simultaneously interpolate policy distributions and furnish closed-form Sobol' sensitivity indices at a negligible computational cost.

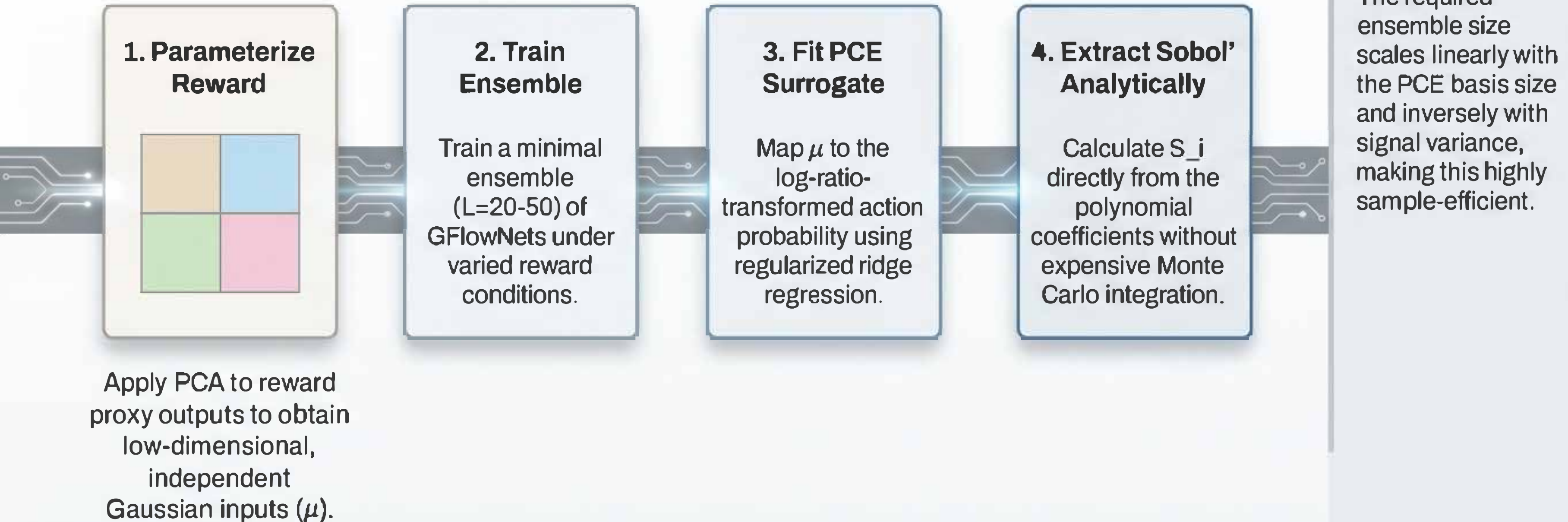
The PCE surrogate acts as a prism, splitting opaque variance into an interpretable spectrum.

Sobol' Indices ($S_i = D_i/D$)

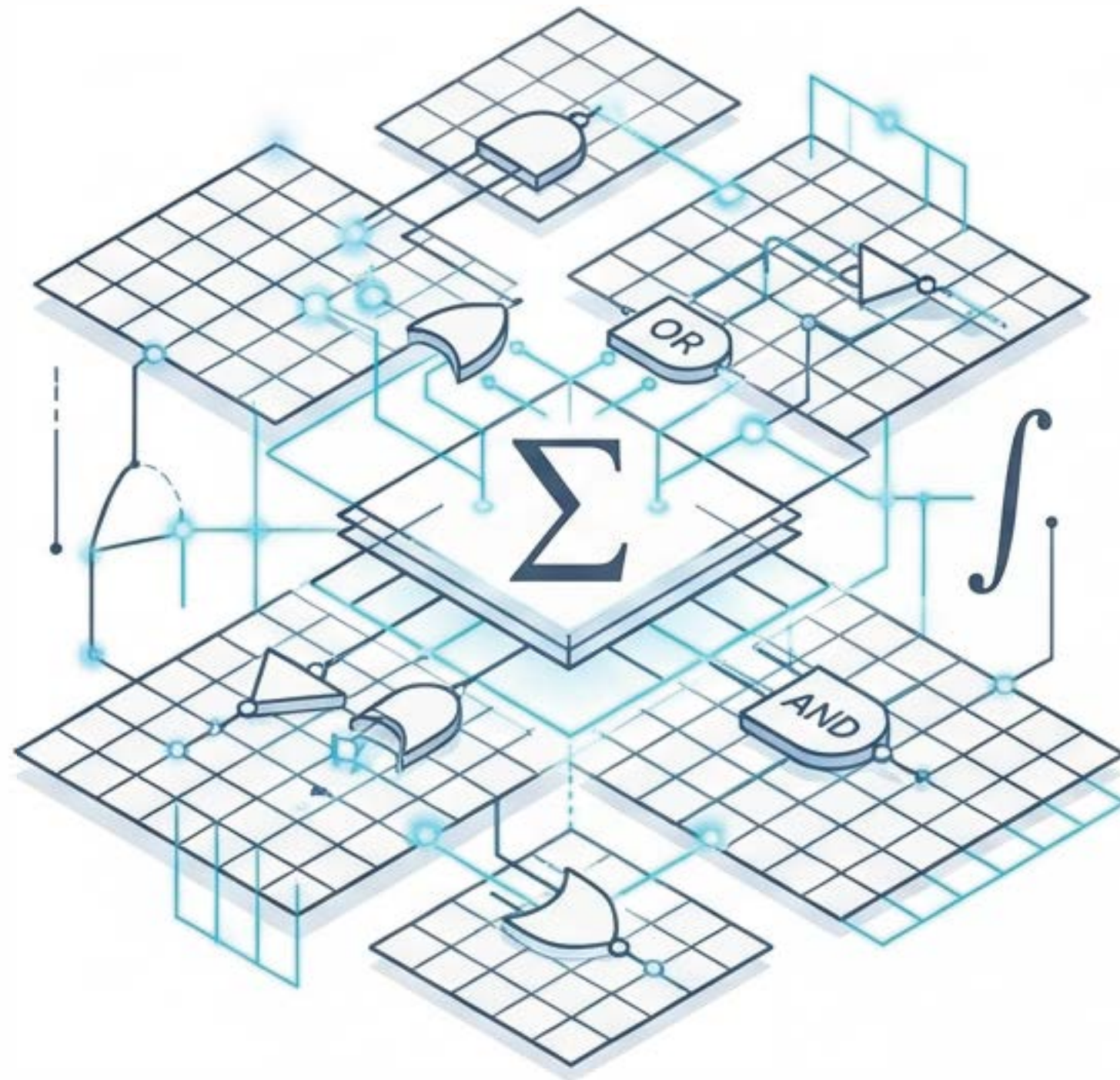


Takeaway: By mapping reward parameterization directly to policy distributions, we isolate the exact drivers of generative decisions.

A four-stage pipeline for **extracting analytical sensitivity**.



Rigorous theoretical guarantees verified via the Lean 4 proof assistant



PCE Convergence Rate

Convergence scales optimally at $O(p^{-s})$ for smoothness s and degree p .



Simplex Preservation

Softmax normalizes correctly; all probability entries remain strictly positive.



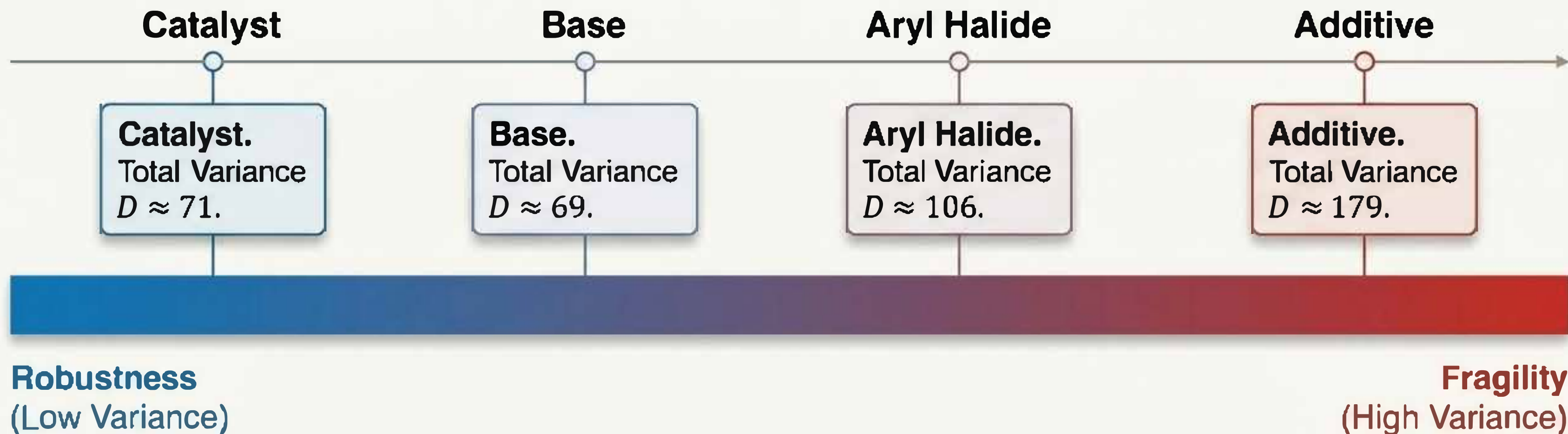
Uncertainty Propagation. An ε -perturbation in reward induces at most $O(L\varepsilon)$ perturbation in the policy.



Sobol' Convergence. Surrogate-estimated indices continuously approach true indices as L_2 error vanishes.

Mathlib formalization confirms constructive bounds and simplex topology

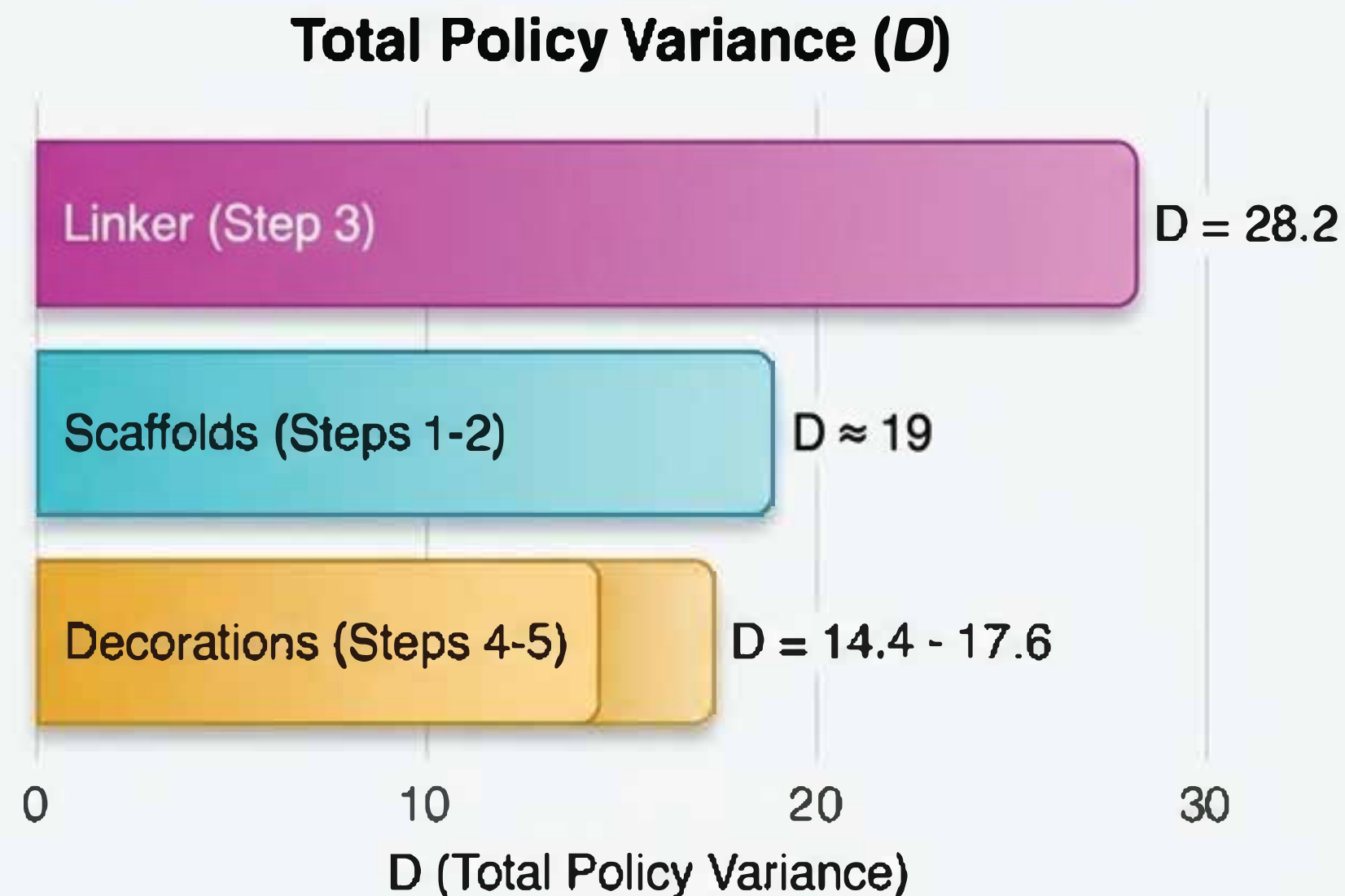
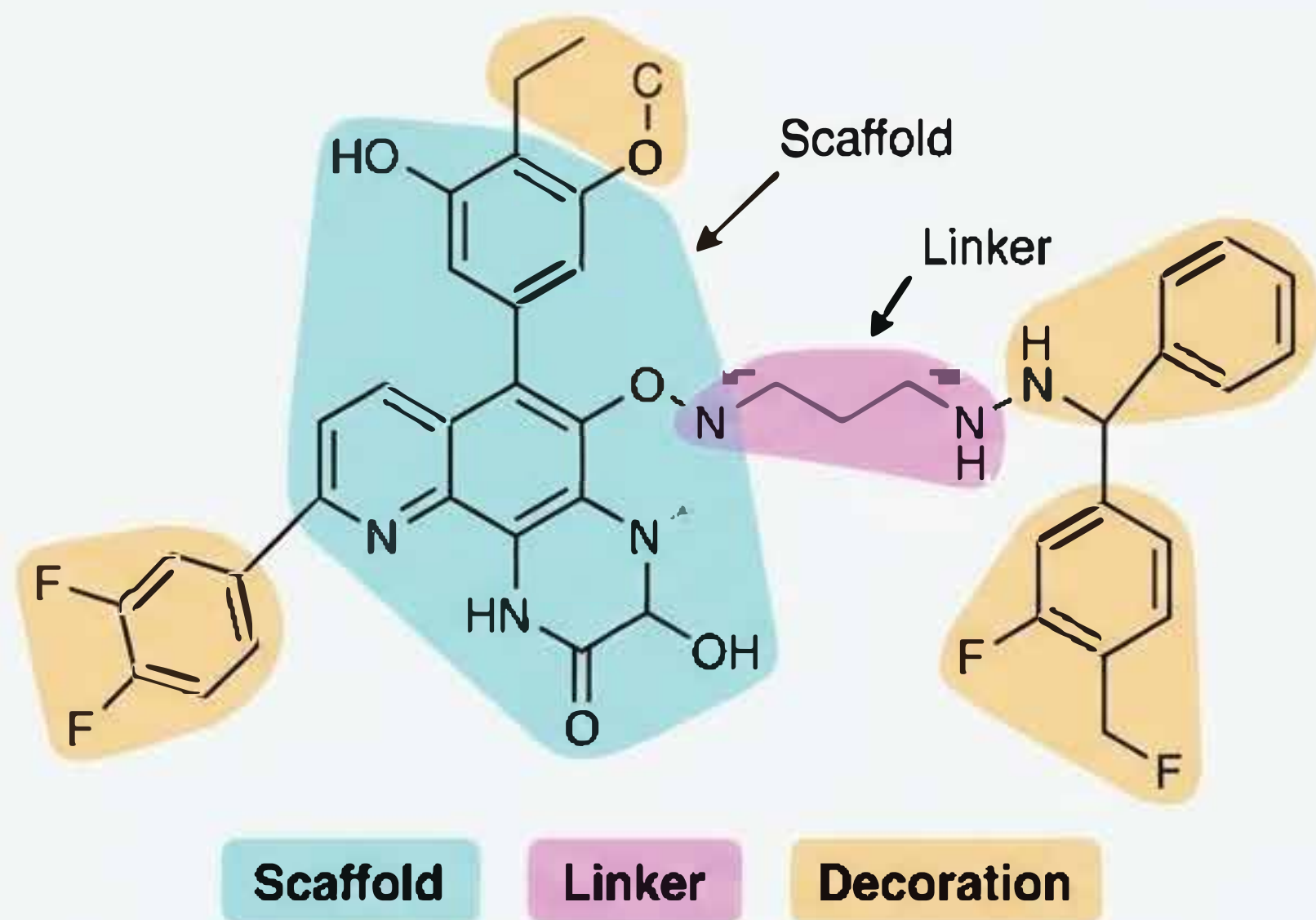
Reaction Optimization: Separating **genuine chemical signals** from **training artifacts**.



The Chuang-Keiser Resolution

Catalyst choices reflect genuine chemical signal, robust to yield-predictor uncertainty. Additive choices carry 2.5x higher variance, flagging them as fragile artifacts highly sensitive to training subset biases.

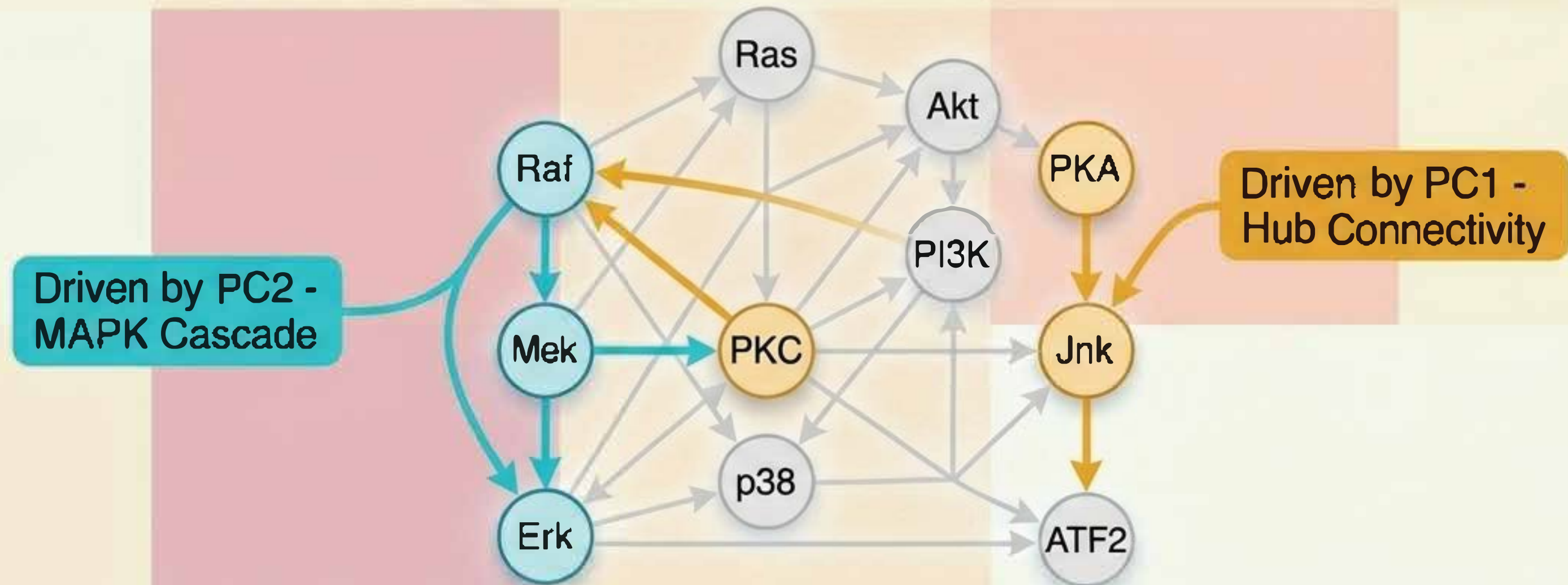
Molecular Design: Reversing medicinal chemistry intuition.



Insight

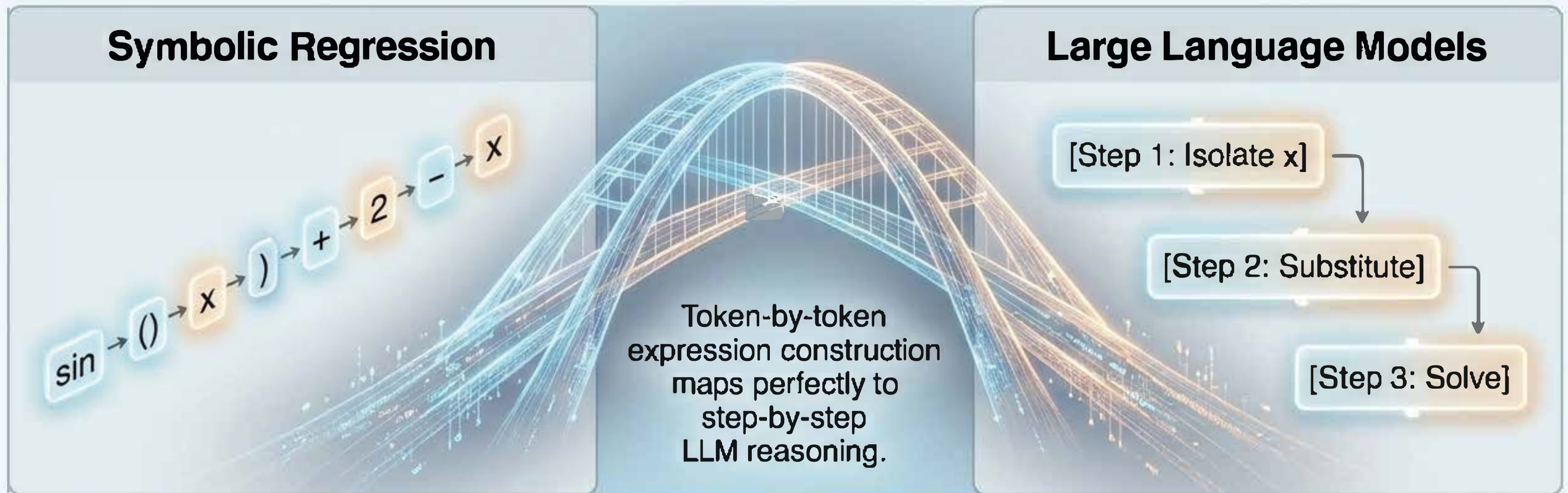
Standard medicinal chemistry assumes scaffolds are robust and decorations are fragile. The PCE surrogate proves the exact opposite in this reward landscape: **linkers are the most fragile** because their geometry mediates all downstream interactions.

Causal Discovery: Pinpointing the **limits** of observational biology.



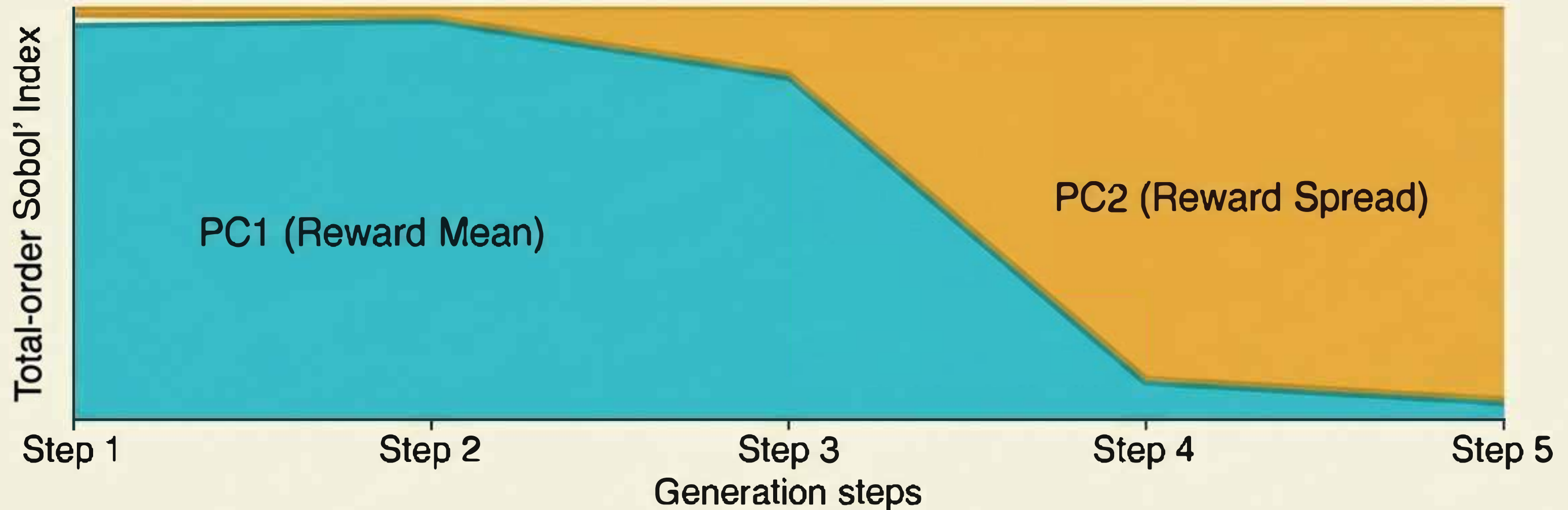
200 observations cannot resolve this 11-node network. However, the Sobol' decomposition provides a precise experimental roadmap: target interventions on the Raf/Mek axis to reduce MAPK uncertainty, or target PKA to reduce hub uncertainty.

Scaling the prism: Generative reasoning and Process Reward Models (PRMs).



Context: Modern LLMs are fine-tuned with GFlowNets using Process Reward Models (PRMs). Because PRMs are trained on limited human preference data, they carry massive epistemic uncertainty. The PCE surrogate directly translates to quantifying LLM reward hacking.

Temporal variance: How reasoning uncertainty evolves across sequence generation.



Insight: At early steps, policy uncertainty is driven by the overall mean level of the PRM (PC1). At later steps, where the model must discriminate between syntactically similar but distinct logical completions, the variance in reward spread (PC2) becomes the dominant driver of uncertainty.

A computational breakthrough.

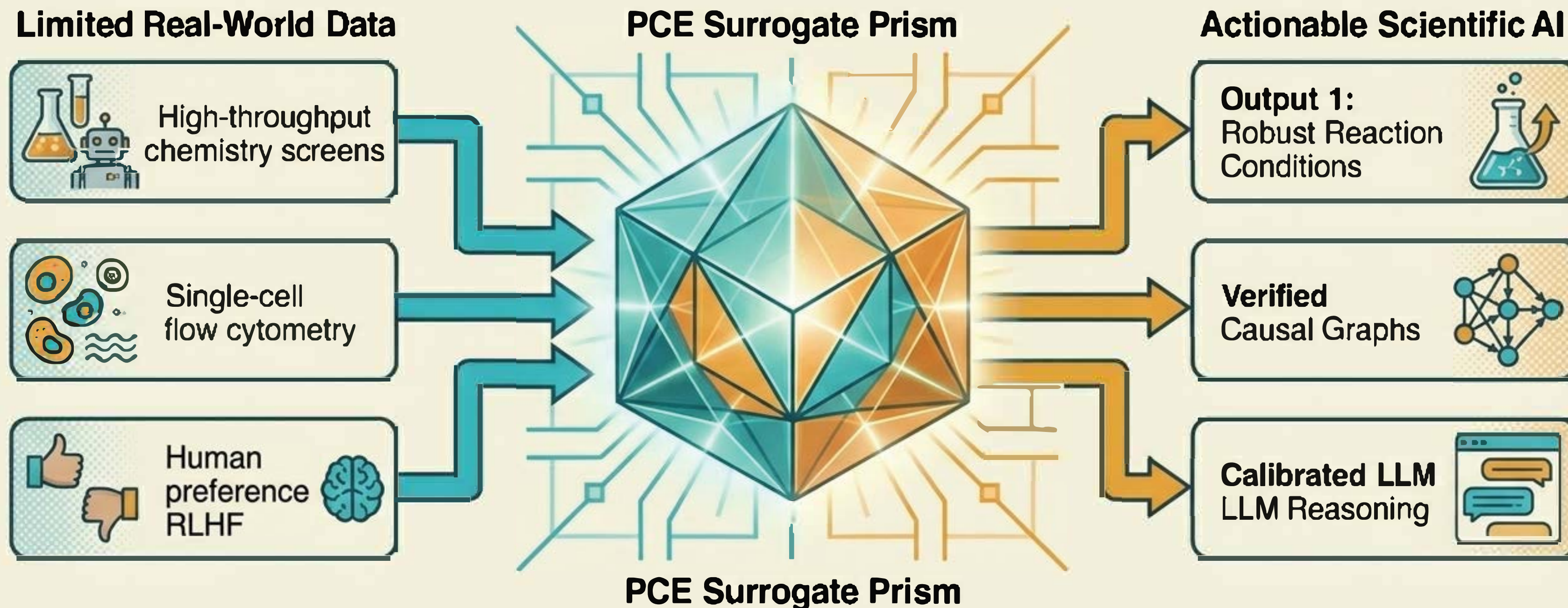
**33,000x
Faster**
over exhaustive retraining.

Computational Cost Comparison

Task	LLM GFlowNet
Ensemble Training Time	~0.5 hours
PCE Fit Time	0.005 seconds
PCE Sample (10k) Time	0.049 seconds

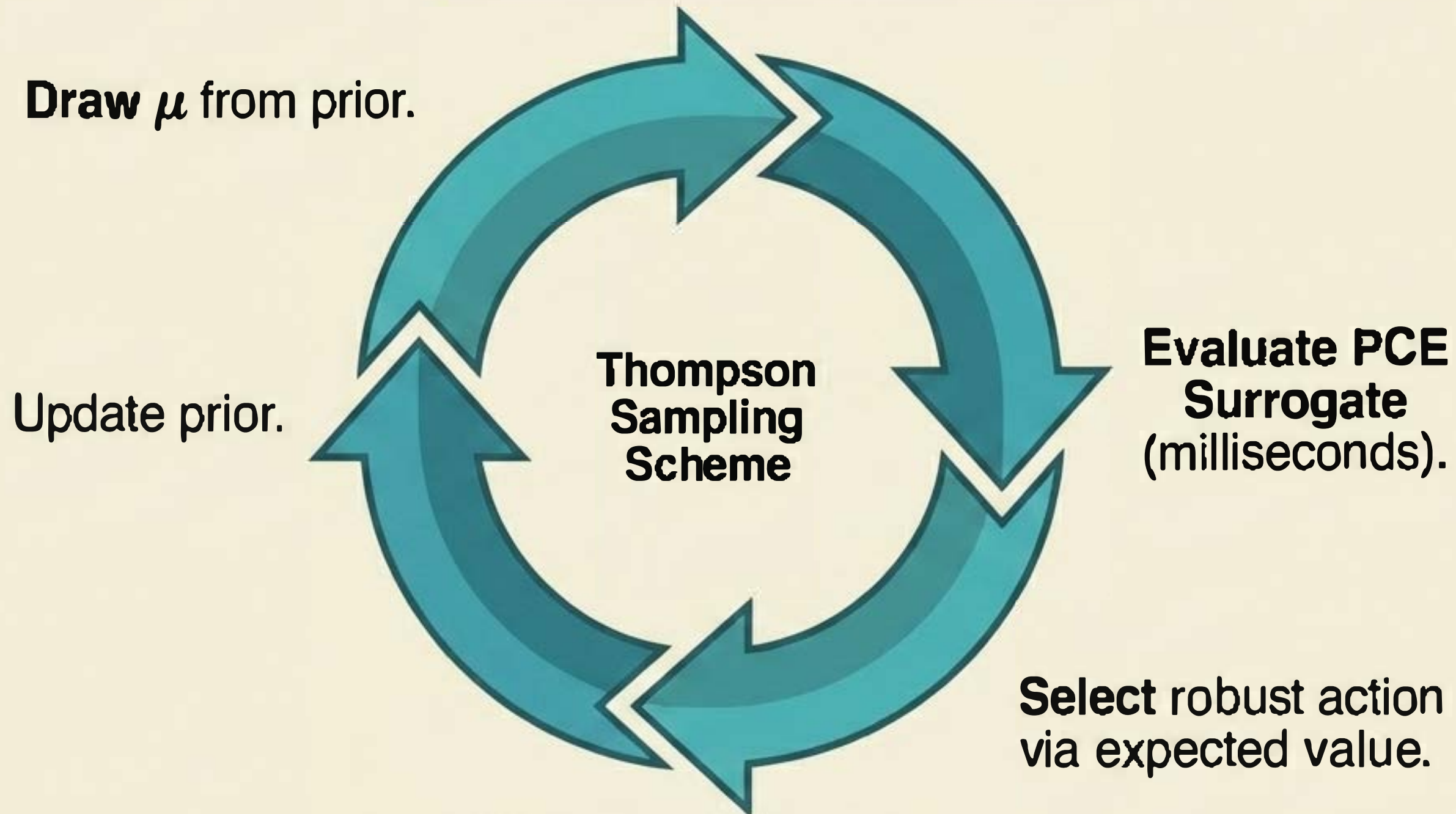
Takeaway: Generating 10,000 surrogate policy samples takes milliseconds. It is virtually free compared to retraining models, enabling real-time, on-the-fly uncertainty quantification without sacrificing accuracy.

A unified architecture for sequential AI discovery.



Regardless of the domain—physics, chemistry, biology, or language—limited data breeds uncertain rewards. The PCE framework provides a single, mathematically verified umbrella to bridge the gap between noisy data and reliable, sequential generation.

From quantifying uncertainty to adaptive generation.



We are moving beyond models that merely *report* uncertainty. By embedding the PCE surrogate directly into the sampling loop, next-generation GFlowNets can **adaptively steer away** from fragile decisions in real-time, autonomously navigating the boundaries of their own ignorance.